

SurVOLLTage

*Évolution et modernisation des techniques de
développement des librairies VOLLT*

Semi-hack-a-thon – Journées OV France

Mai 2025

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Introduction

The background of the slide is a dark blue gradient. It is decorated with numerous glowing blue lines of varying thickness and length, some of which are curved and others straight. Scattered throughout the background are many small, bright blue dots or particles, some of which are clustered together, creating a sense of depth and movement. The overall effect is reminiscent of a cosmic or digital space.

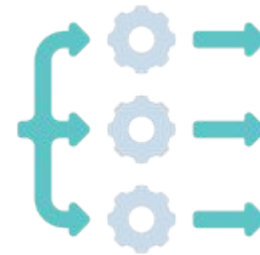
Développements à venir pour VOLLT

	Usage 	Interne 
 PROGRESS TAP-1.1 + ADQL-2.1	Publication dans Maven Central	 PROGRESS Support des tableaux
Générer entrée Registry	 PROGRESS Doc. par l'exemple	LogBack et non log maison
Détecter le(s) DataModel(s)	Fournir Docker	Indépendance HTTPServlet
DataLink + <LINK>	Job List en BdD	ExecutorService dans UWS
ConeSearch	Rechargement à chaud	
UPLOAD depuis VOSpace	Stats en BdD	
	OpenAPI	
...	...	...



Ouverture à la collaboration

Plus on est de fous, plus on rit...



Avec GitHub, mais pas que...

GitHub

- Issues
- PullRequests
- Forks

Système d'extension dans VOLLT



Des suggestions ?





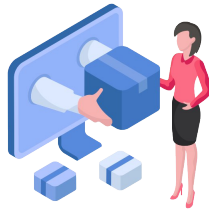
Modernisation de l'environnement

Gradle, à l'aide !



Gradle

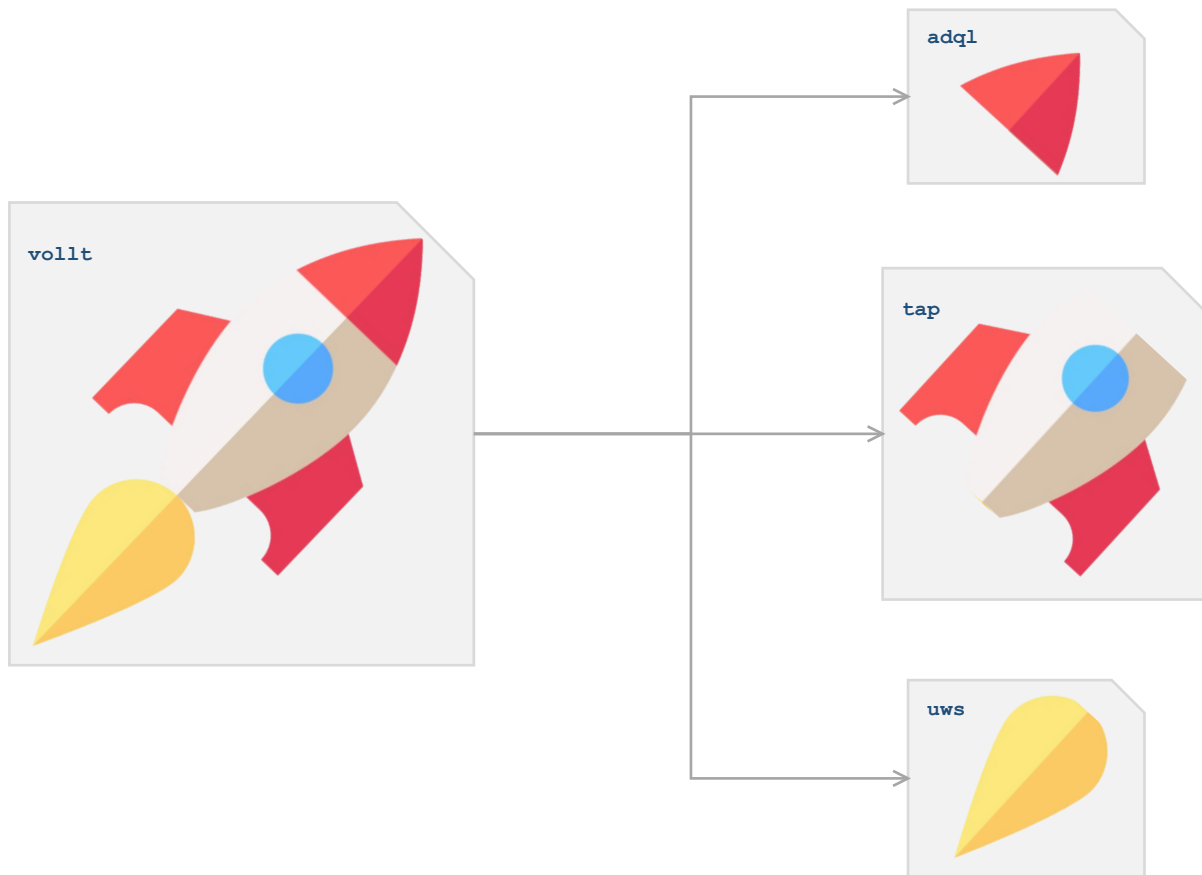
Groovy/Kotlin



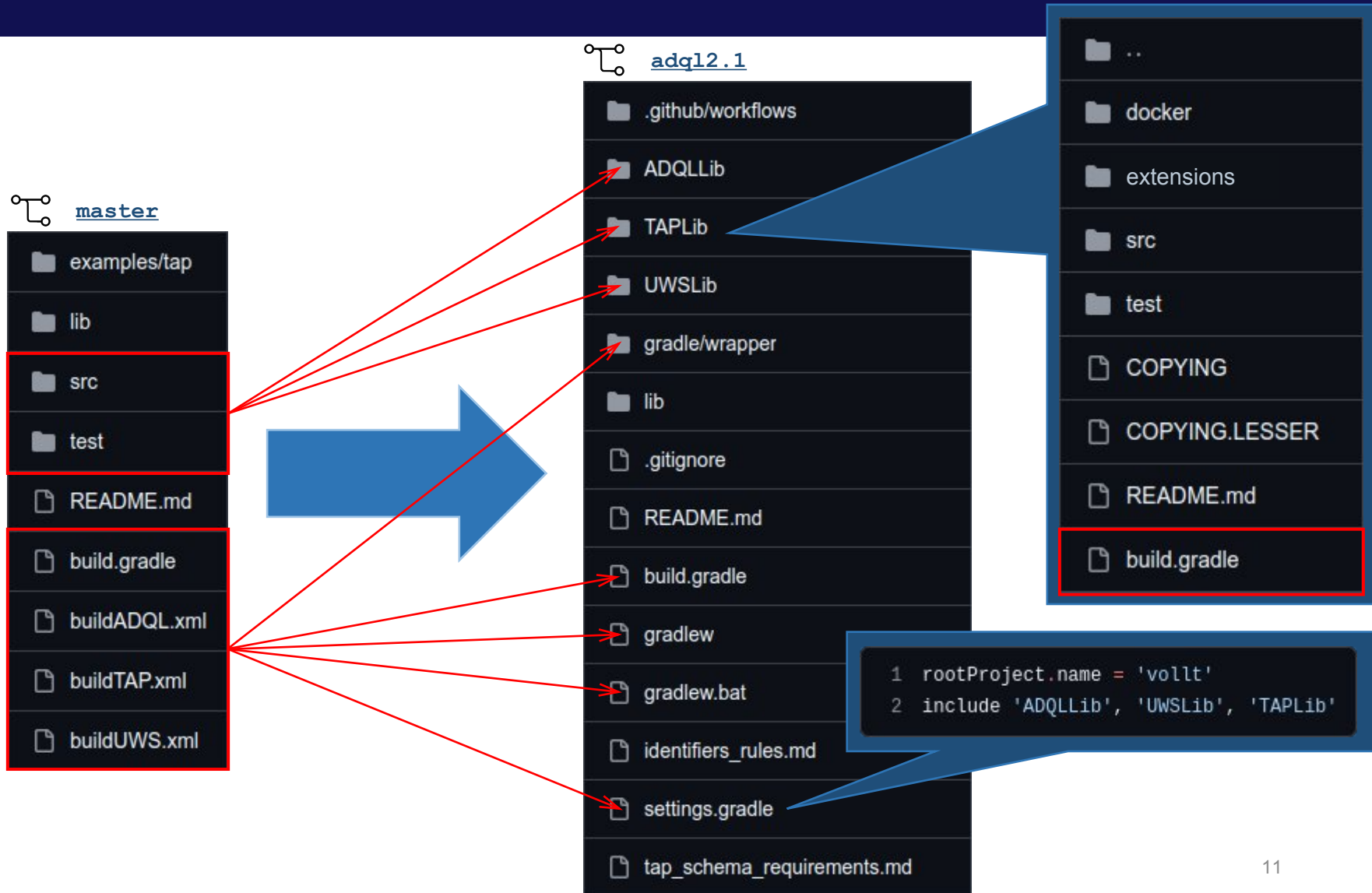
= **Maven**[™]

XML

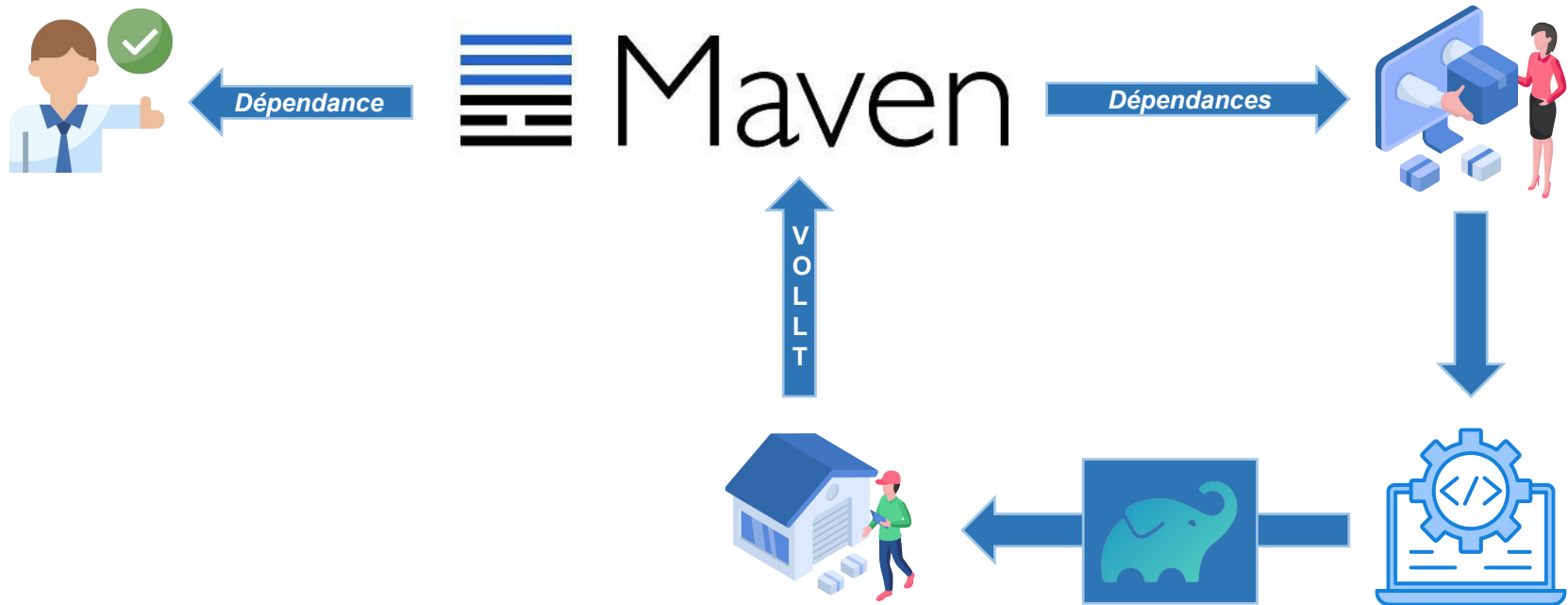
1 projet = 3 sous-projets



Chacun gère ses affaires



C'est donnant-donnant



C'es



```
publishing {
    publications {
        maven(MavenPublication) {
            groupId = 'fr.unistra.cds' //FIXME - is this what is wanted?
            artifactId = 'ADQLlib'
            version = '2.0-SNAPSHOT'

            from components.java
            pom {
                name = 'ADQLlib'
                description = 'Java library to parse, analyze and transform ADQL queries.'
                url = 'http://cdsportal.u-strasbg.fr/adqltuto/'

                licenses {
                    license {
                        name = 'GNU Lesser General Public License version 3'
                        url = 'https://opensource.org/license/lgpl-3-0'
                    }
                }
                developers {
                    developer {
                        id = 'gmantele'
                        name = 'Grégory Mantelet'
                        email = 'gregory.mantelet@astro.u-strasbg.fr'
                    }
                }
                scm {
                    connection = 'scm:git:git://github.com/gmantele/volllt.git'
                    developerConnection = 'scm:git:ssh://github.com/gmantele/volllt.git'
                    url = 'http://example.com/my-library/'
                }
            }
        }
    }
}
```





Des questions ?
Des suggestions ?

The background of the slide is a dark blue field filled with abstract, glowing elements. There are numerous thin, wavy lines in shades of light blue and white, some of which are thicker and more prominent. Scattered throughout the field are many small, bright blue dots or particles, some of which are clustered together. A horizontal band of slightly larger, more concentrated blue particles runs across the lower middle of the image. The overall effect is one of dynamic energy and complexity.

Évolution de développement

Une IDEA en Eclipse une autre

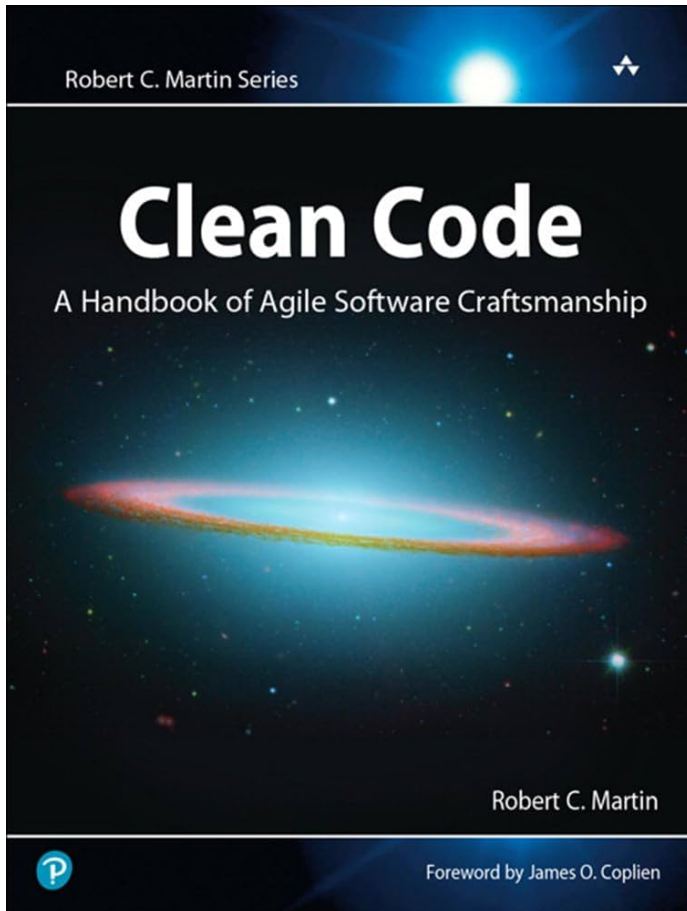


IntelliJ IDEA



- Plus rapide (subjectif)
- Mise à jour auto. intégrée
- Bonne complétion intelligente de code (avec ou sans IA)
- Plein d'extensions
- Version payante :
 - Couplage avec serveur applicatif (Tomcat, ...)
 - Client BdD
 - Visualisation des classes en diagramme UML
 - ...

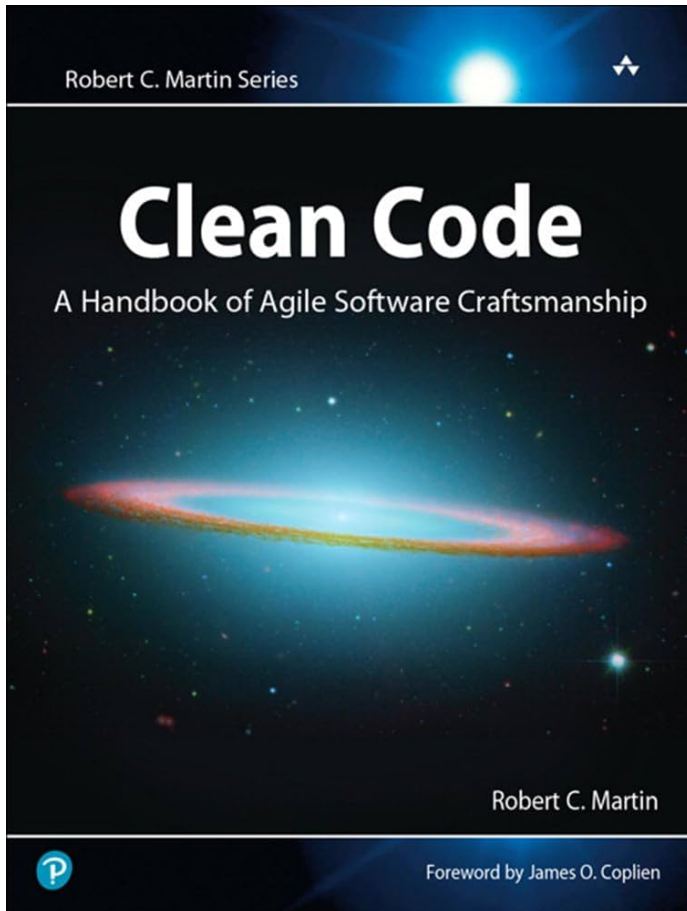
Dresscode ? Juste un code propre ! - 1/5



Réduction de la complexité (cognitive) du code

- Fonctions
 - courtes (~ 2-3 lignes)
 - qui ne fait qu'une seule chose et le fait bien
 - Peu de paramètres (max 3)
 - Pas de paramètre « flag »
- 1 classe = 1 responsabilité
- Utiliser des concepts bien établis et connus (ex : design pattern)
- ...

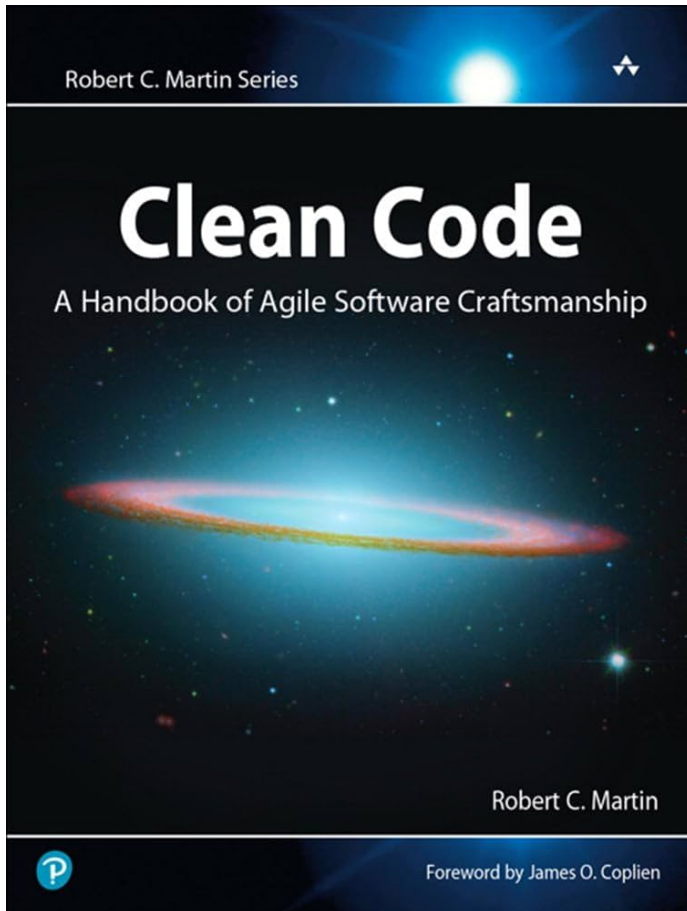
Dresscode ? Juste un code propre ! - 2/5



Éviter les répétitions

- De code
- De constantes

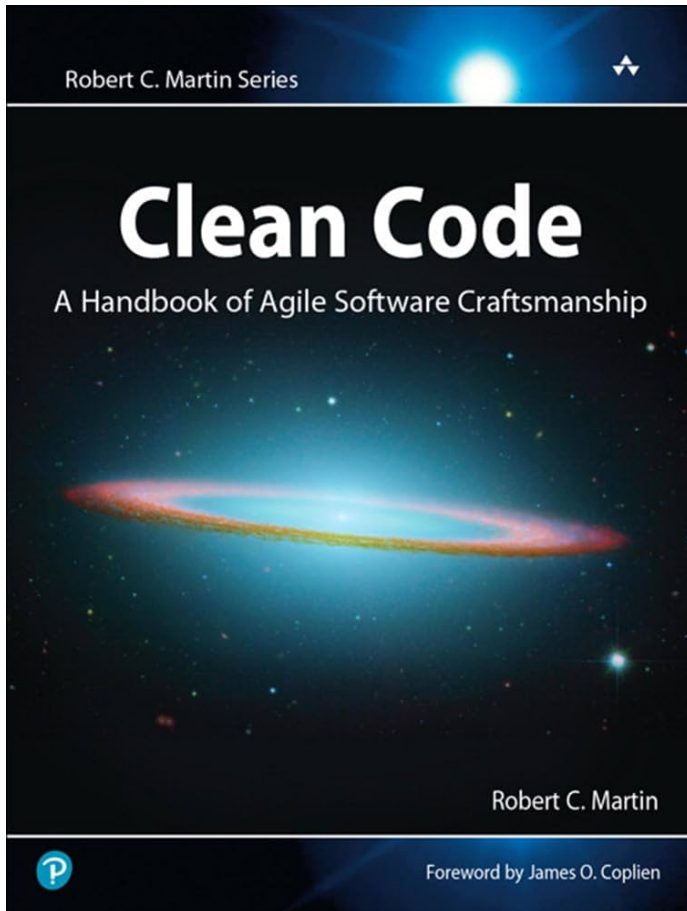
Dresscode ? Juste un code propre ! - 3/5



Lisibilité du code

- Lisible comme un texte
- Noms variables/fonctions/classes explicites
- Fonctions ordonnées par appel
- Éviter les conditions négatives
- N'utiliser des commentaires que quand c'est vraiment utile ; pollution visuelle
 - 1 commentaire pour 1 intention
 - 1 avertissement pas évident
- ...

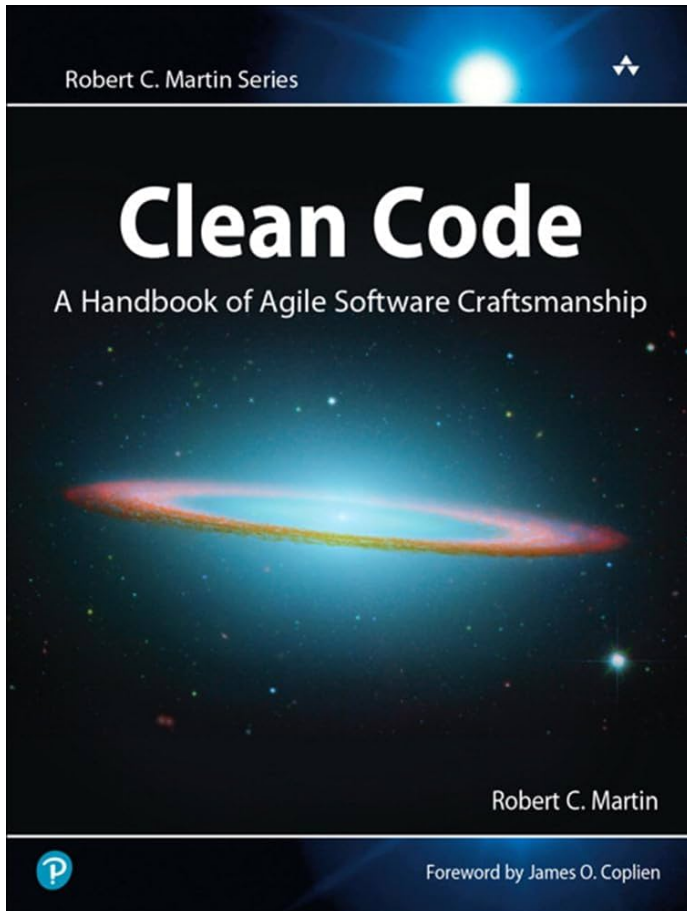
Dresscode ? Juste un code propre ! - 4/5



Ne pas utiliser/retourner de `null`

- Facilité $\Rightarrow$ `NullPointerException`
- Solutions possibles:
 - Optional
 - Liste/Tableau vide
 - Exception
 - ...

Dresscode ? Juste un code propre ! - 5/5



Tests

- Lisibles et courts
- Couverture raisonnable et maximale
- Nom lisible
(ex. dans VOLLT:
`NomFonction_Should...._When...`)
- Structure explicite d'un test
(ex. dans VOLLT: `Given > When > Then`)
- ...

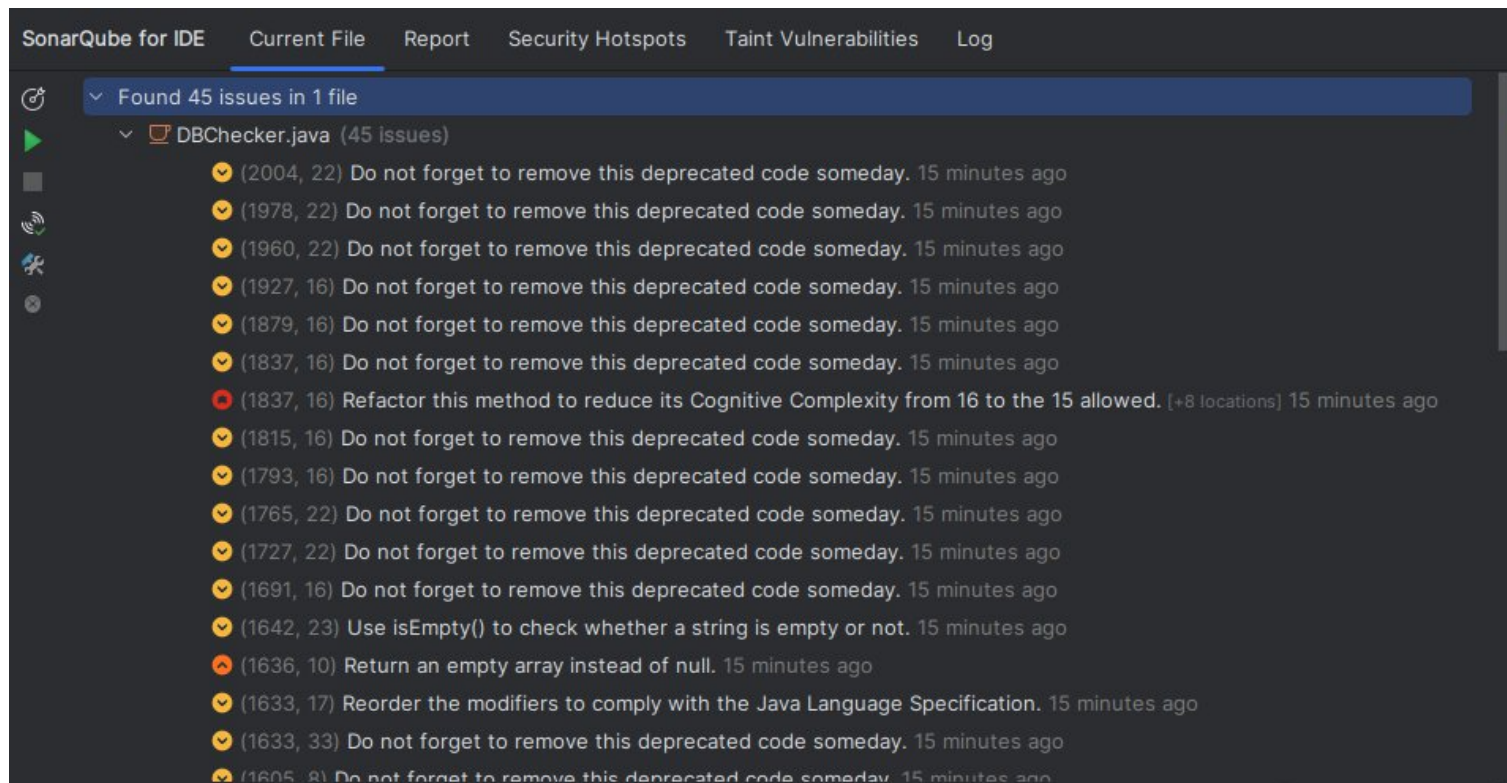
Un méchant inspecteur dans mon IDE 1/5



Un méchant inspecteur dans mon IDE 2/5



SonarQube
ide



Un méchant inspecteur dans mon IDE 3/5



```
resolveTables(final ADQLQuery query, final Deque<CheckContext> contextList, final Unre  
Context c  
dler sHand  
ne existen  
new Search  
earch(quer  
ject resul  
QLTable ta
```

Refactor this method to reduce its Cognitive Complexity from 29 to the 15 allowed. ⋮
SonarQube: Show issue locations Alt+Shift+Enter More actions... Alt+Enter

© adql.db.DBChecker

```
protected void resolveTables(  
    @NotNull ADQLQuery query,  
    @NotNull Deque<CheckContext> contextList,  
    UnresolvedIdentifiersException errors  
)
```

62 lignes et 3 niveaux d'imbrication

Un méchant inspecteur dans mon IDE 4/5

Cognitive Complexity of methods should not be too high

Adaptability issue | Not focused

Maintainability 

java:S3776

[Learn more](#)

This rule raises an issue when the code cognitive complexity of a function is above a certain threshold.

[Why is this an issue?](#)

[How can I fix it?](#)

[More Info](#)

Cognitive Complexity is a measure of how hard it is to understand the control flow of a unit of code. Code with high cognitive complexity is hard to read, understand, test, and modify.

As a rule of thumb, high cognitive complexity is a sign that the code should be refactored into smaller, easier-to-manage pieces.

Which syntax in code does impact cognitive complexity score?

Here are the core concepts:

- **Cognitive complexity is incremented each time the code breaks the normal linear reading flow.**
This concerns, for example, loop structures, conditionals, catches, switches, jumps to labels, and conditions mixing multiple operators.
- **Each nesting level increases complexity.**
During code reading, the deeper you go through nested layers, the harder it becomes to keep the context in mind.
- **Method calls are free**
A well-picked method name is a summary of multiple lines of code. A reader can first explore a high-level view of what the code is performing then go deeper and deeper by looking at called functions content.
Note: This does not apply to recursive calls, those will increment cognitive score.

The method of computation is fully detailed in the pdf linked in the resources.

Exceptions

`equals` and `hashCode` methods are ignored because they might be automatically generated and might end up being difficult to understand, especially in the presence of many fields.

Un méchant inspecteur dans mon IDE 5/5



```
if (!context.cteTables.add(dbTable))
    erro
(ParseExc
```

Dereference of 'context' may produce 'NullPointerException' :
Assert 'context != null' Alt+Shift+Enter More actions... Alt+Enter

```
// set the corresponding tables among the list of resolved tables:
//wildcard.setAdqlTable(mapTables.get(dbTable));
wildcard.getAdql
atch(ParseExcepti
errors.addExcept
```

This block of commented-out lines of code should be removed. :
SonarQube: Show rule description 'java:S125' Alt+Shift+Enter More actions... Alt+Enter

```
if (table.isSubQuery() || table.hasAlias()) {
    if (!context.cteTables.add(dbTable))
}
atch(Pa
```

Merge this if statement with the enclosing one. :
SonarQube: Merge this if statement with the enclosing one Alt+Shift+Enter More actions... Alt+Enter

```
if (dbTable == null)
    dbTable =
// wrap this t
if (dbTable !=
```

Condition 'dbTable == null' is always 'true' :
Unwrap 'if' statement Alt+Shift+Enter More actions...

```
@Deprecated Grégory Mantelet +1
protected final static String[] specialSort(final Collection
// Nothing to do if the array is NULL:
if (items == null)
    return null;
// Keep only
```

SonarQube: Return an empty array instead of null. 26



Des questions ?
Des suggestions ?

Conclusion

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Récap



IDE

Idea IntelliJ + SonarQube



Développement

CleanCode et bonnes pratiques



Qualité

tests de couverture plus important



Collaboratif

GitHub et visio (hands-on ?)



Satisfaction

personnelle et des utilisateurs

Merci

